# Ayoub Bougatia

Ben Guerir / Casablanca, Morocco

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## PROFESSIONAL SUMMARY

1337 Coding School (42 Network) student and Industrial Digitalization graduate with hands-on experience in C/C++ systems programming, non-blocking networking, containerized deployments, and infrastructure security. Seeking a 2027 PFE internship in Software Engineering / DevOps.

## TECHNICAL SKILLS

**Programming:** C, C++98, Bash / Shell; familiarity with Python and SQL
**DevOps & Security:** Linux, Docker, Docker Compose, NGINX, HTTPS/TLS, HashiCorp Vault (AppRole), Git
**Systems & Networking:** TCP/IP, POSIX sockets, `poll()`, multithreading, mutexes, GDB, Valgrind
**Platform Operations:** PostgreSQL service integration and backups, reverse proxying, service health checks

## SELECTED TECHNICAL PROJECTS

**ft_transcendence – Real-Time Web Platform** Docker, NGINX, Vault, PostgreSQL
*DevOps & Security Lead | 5-person team* *2026*

- Owned infrastructure and security integration for a team-built web application; deployed frontend, FastAPI backend, PostgreSQL, NGINX, and Vault using Docker Compose.
- Configured HTTPS reverse proxying and Vault AppRole secret access with least-privilege policies; validated database connectivity, service readiness, and PostgreSQL backup integrity.

**ft_irc – IRC Network Server** C++98, TCP/IP, POSIX sockets
*Networking & Server Foundation | 3-person team* *2026*

- Implemented the TCP server networking layer in C++98, including socket setup (`socket`, `bind`, `listen`, `accept`) and client connection management.
- Built non-blocking I/O multiplexing with `poll()` to handle concurrent client reads, writes, and disconnects within an event-driven server.

**Inception – Containerized Service Infrastructure** Docker, Compose, Debian, TLS
*Infrastructure Project* *2026*

- Built a multi-container environment for NGINX, WordPress/PHP-FPM, and MariaDB using custom Dockerfiles, internal networking, and persistent volumes.
- Configured TLS termination and container restart behavior; tested communication between services and persistence across restarts.

**Minishell & Philosophers – UNIX Systems Programming** C, POSIX, pthreads
*Minishell: 2-person team | Philosophers: individual* *2025*

- Co-developed a UNIX-like shell supporting pipelines, redirections, environment expansion, and signal handling.
- Used threads and mutexes to coordinate shared resources in a dining-philosophers concurrency simulation.

## PROFESSIONAL EXPERIENCE

**OCP Group – Mining Production** Ben Guerir, Morocco
*Industrial Digitalization Intern (Licence Professionnelle PFE)* *Apr – Jun 2025*

- Analyzed operating processes, sensor instrumentation, and safety constraints for the proposed automation of an industrial stacker-reclaimer machine.
- Designed a proposed machine-to-control-room network architecture and prepared technical specifications and a techno-economic feasibility study.

## EDUCATION

**1337 Coding School (42 Network / UM6P)** Ben Guerir, Morocco
*Software Engineering – Common Core Completed* *2024 – Present*

**Green Tech Institute (UM6P)** Ben Guerir, Morocco
*Licence Professionnelle – Industrial Operations & Digitalization* *2022 – 2025*

## LANGUAGES & LEADERSHIP

**Languages:** English (IELTS C1), French (DELF B2), Arabic (Native)
**Leadership:** Events Manager, Digital Innovation & Technology Club (UM6P)